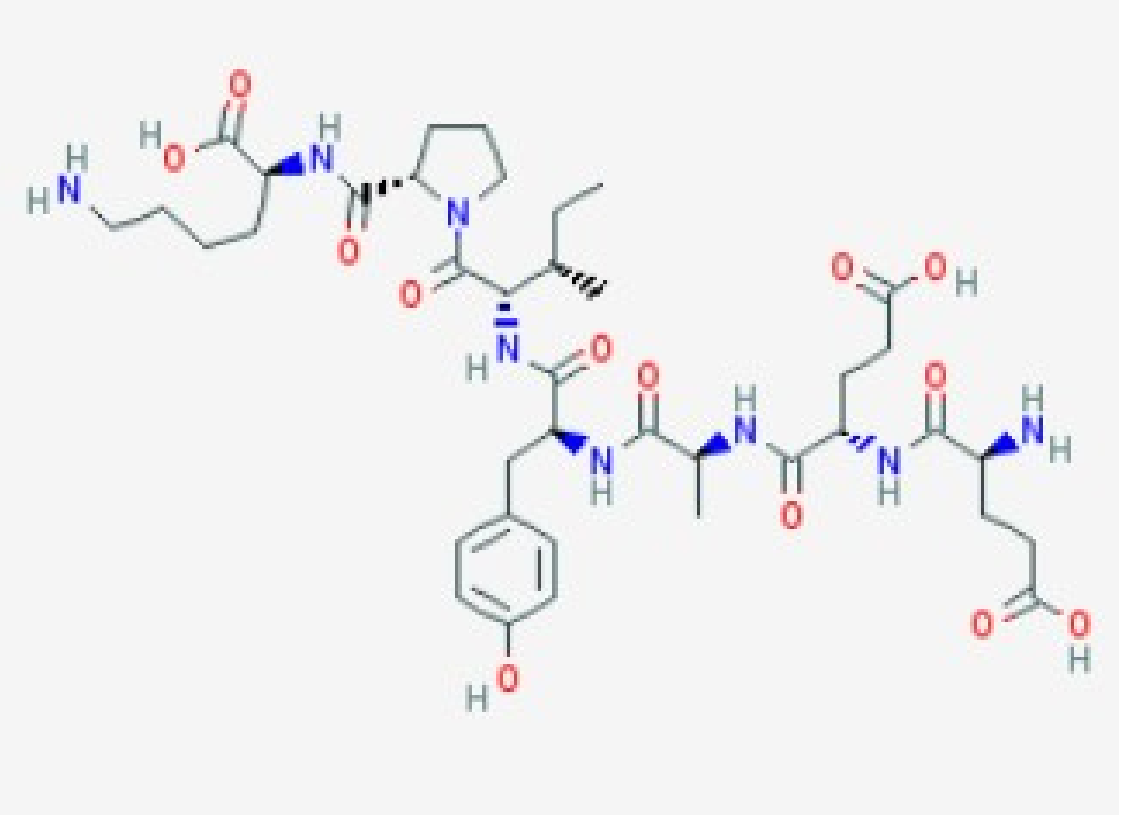
# Growth Hormone



Production, Effects, Regulation and Disease

# Growth Hormone

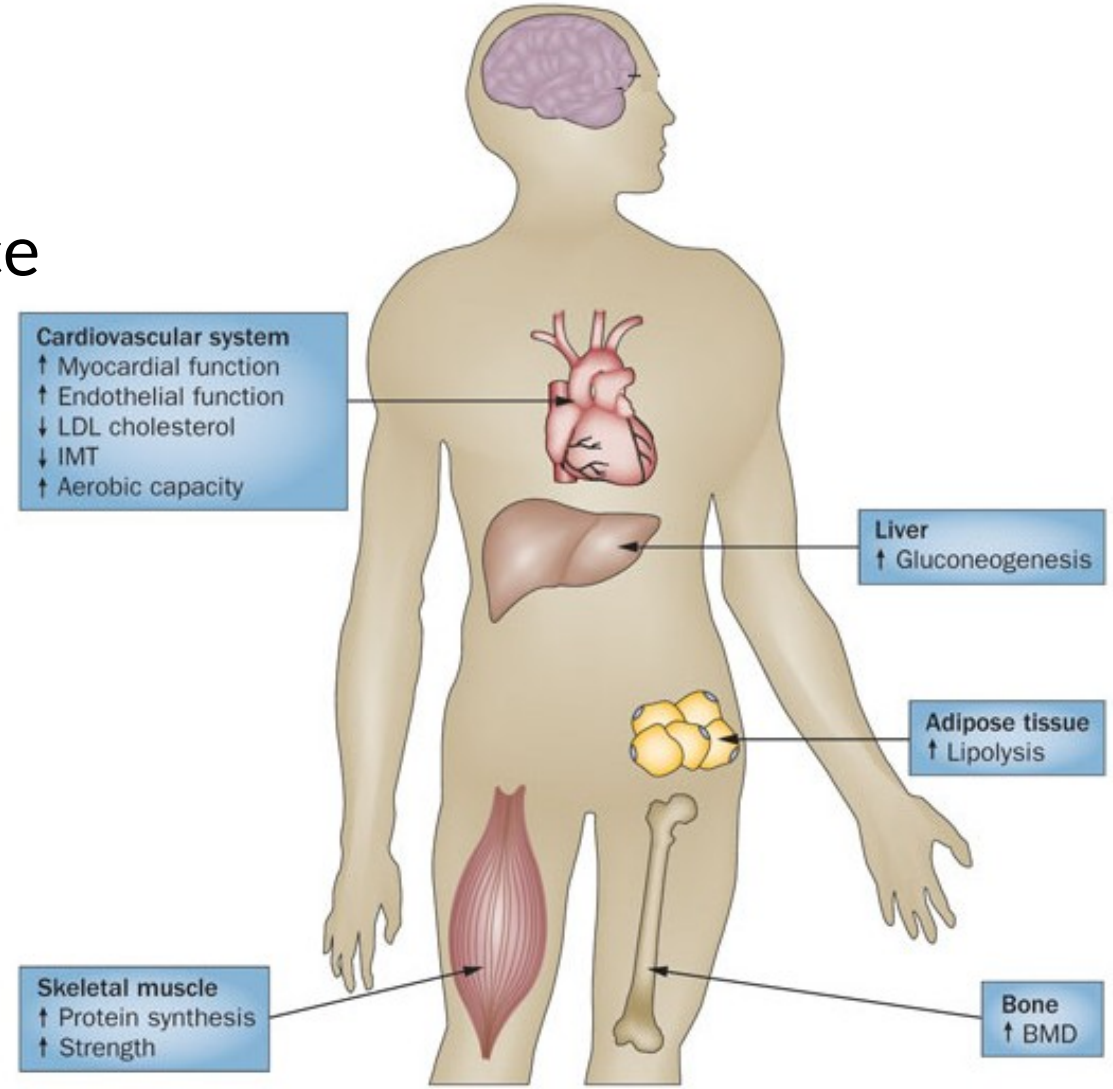
- Also called *somatotropin*
- Peptide Hormone (protein/amine group)
- Produced by Anterior Pituitary and released into blood
- Variety of effects related to growth and development



*Learning Aim: Explain where Growth Hormone is produced.*

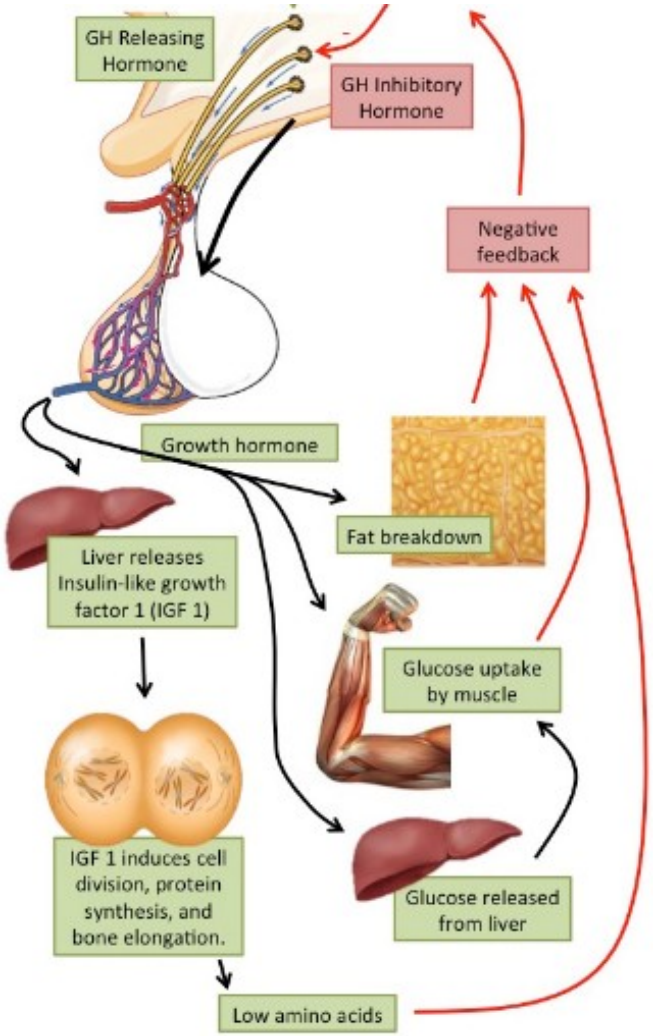
# Effects of Growth Hormone

- Anabolic:  “builds up” tissues
- Acts with a specific receptor on cell surface
- Increases:
  - Height (bone and muscle growth) in children and adolescents
  - protein synthesis
  - metabolism of fats
  - Muscle mass
  - Bone mineralization and density
  - Growth of internal organs (but not brain)
- Acts on liver to increase blood glucose:
  - Reduces uptake of glucose by liver
  - Increases gluconeogenesis




*Learning Aim:  List the effects on the body of Growth Hormone.*

# Growth Hormone Regulation




- Sex hormone interactions during puberty
- Deep sleep
- Nutrition
- Exercise
- Action of certain neurotransmitters

**Hypothalamus**

Produces and releases GHRF (Growth Hormone Releasing Factor)

Produces and releases GHIF (Growth Hormone Inhibiting Factor) also called somatostatin

Travels through blood vessels in infundibulum

**Anterior Pituitary**

Produces and releases GH (Growth Hormone) also known as somatotropin

Travels in systemic circulation

**Body Tissues**

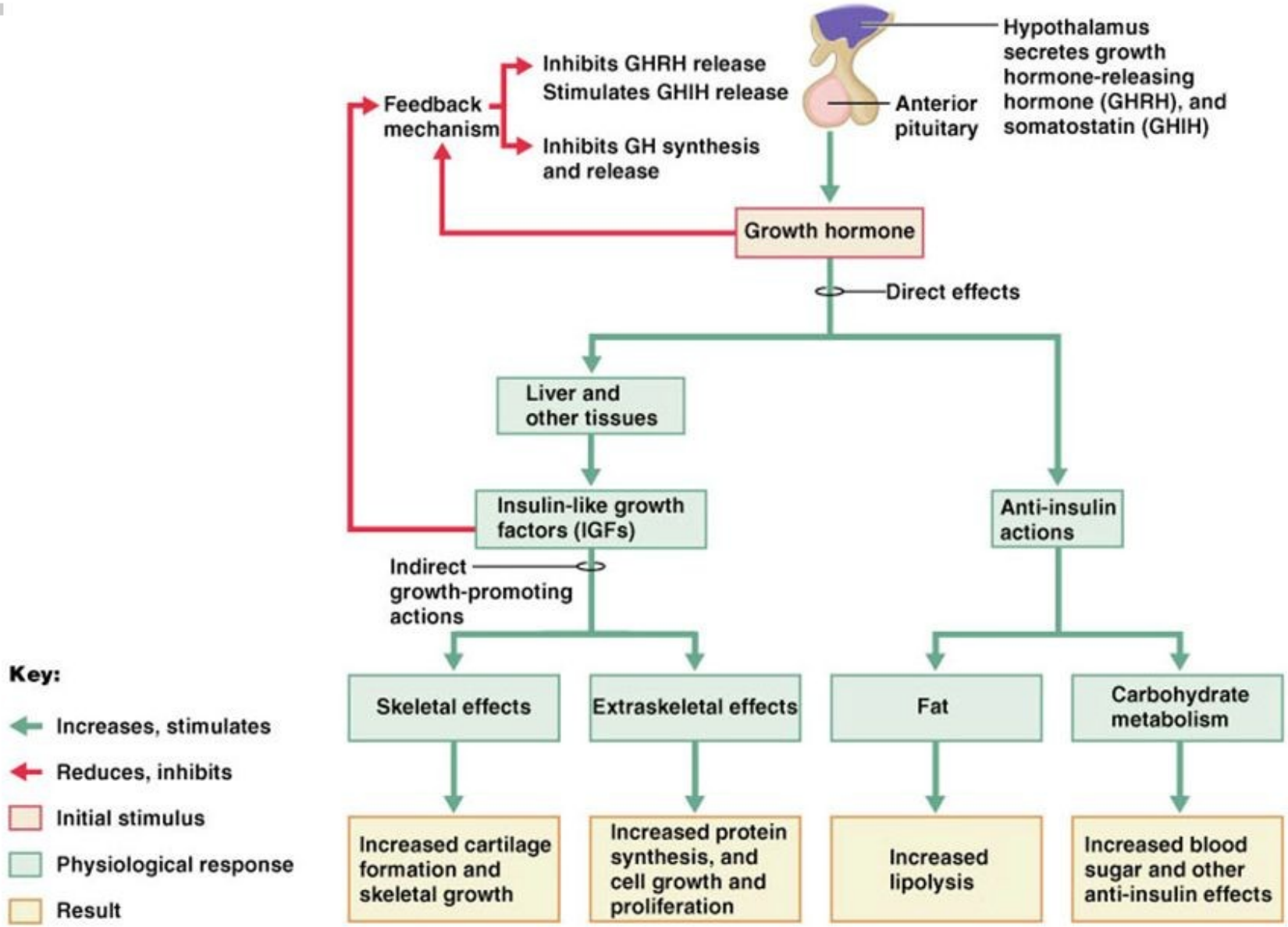
- Liver produces factors (somatomedins) that induce cell division, protein synthesis from amino acids and bone elongation
- Liver converts glycogen to glucose and releases it into bloodstream
- Glucose is taken up by skeletal muscle
- Adipose tissues break down fat, increasing fatty acids in blood stream.

**Negative Feedback**

Effects of GH on body stimulates release of GHIF from hypothalamus which inhibits release of GH from Anterior Pituitary

*Learning Aim: Describe in in detail how Growth Hormone is regulated.*

# Metabolic Action of Growth Hormone




*Learning Aim: Describe in in detail how Growth Hormone is regulated.*

# Growth Hormone Imbalance and Disease

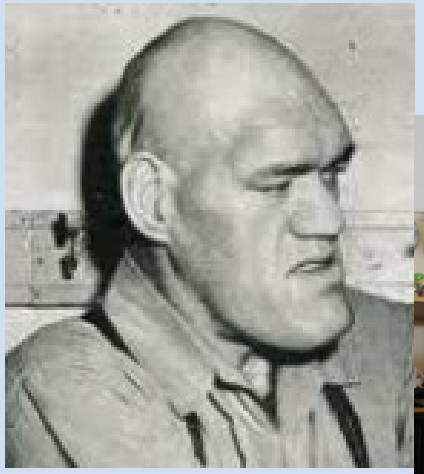
- Problems with Growth Hormone production affect homeostasis, causing disease.

- ***Growth Hormone Hypersecretion***: too much growth hormone.

- ***Growth Hormone Deficiency***:  not enough growth hormone.



# Growth Hormone Hypersecretion

| Common Cause(s) | Symptoms | Diagnosis | Treatment |
|---|---|---|---|
| Commonly: Benign tumours on anterior pituitary gland causing oversecretion of Growth Hormone.<br><br>Occasionally: as a result of inherited diseases or chromosomal abnormalities | Adults: *Acromegaly*<br>• Larger than proportion hands and feet<br>• Thick toes and fingers<br>• Prominent jaw and forehead<br>• Coarse facial features<br>• Sweating<br>• Headaches<br>• Weakness/fatigue<br>• Insomnia<br><br>Children: *Gigantism*<br>• As above, but greater than normal height also. | Blood test for Insulin-Like Growth Factor<br><br>Glucose tolerance test<br><br>MRI to confirm presence of pituitary tumour(s)<br><br>   | • Targeted radiation therapy to shrink tumours<br><br>• Microsurgery to remove tumours<br><br>• Somatostatin medication to inhibit GH release |

*Learning Aim: Discuss the disorders caused by Growth Hormone hyper- and hypo-secretion*

# Growth Hormone Deficiency

| Common causes | Symptoms | Diagnosis | Treatment |
|---|---|---|---|
| Often no cause can be established in children<br><br>Can go together with other pituitary diseases<br><br>Childhood onset may be due to:<br>• Heritable gene mutations<br>• Congenital disease or malformation<br><br>Adult onset may be due to pituitary damage from<br>• Radiation<br>• Surgery<br>• Autoimmune attack of pituitary | Childhood onset:<br>• Short stature – may be proportional or non-proportional.<br>• Slow growth rate.<br>• Slower muscle development.<br>• Unusual facial features – prominent forehead.<br><br>Adult onset:<br>• Decrease in muscle and bone mass<br>• Heart problems<br>• Baldness (men)<br>• Decreased energy and strength | Blood test to look for abnormally low growth hormone levels.<br><br>  | Treatment with injections of growth hormone.<br>• Recombinant gene technology<br>• Gene to produce growth hormone spliced into bacteria<br>• Genetically engineered growth hormone produced by bacteria<br><br>(more on this later) |

*Learning Aim: Discuss the disorders caused by Growth Hormone hyper- and hypo-secretion*